Compression and Cost in High-Resolution X-Ray CT

Jeppe Klitgaard[a]

[a]*Center for Quantification of Imaging Data from MAX IV,*
Technical University of Denmark,
Kongens Lyngby, Denmark

E-Mail: s250250@dtu.dk

Advances in X-ray detector hardware have led to a substantial increase in matrix sizes, which now commonly reach 4k × 4k, with the resulting canonical X-ray Computed Tomography (CT) reconstruction volumes exceeding 100 GB for 16-bit voxels. This places considerable strain on imaging facilities and their users, driven by the storage costs of high-resolution X-ray imaging.

The compression ratios (CR), compression and decompression times, and quality loss are determined for in-memory compression tests across a large number of lossless and lossy compression schemes, including JPEG-LS and the recent JPEG XL standard. Relevant subsets of the tests are performed in TIFF, DICOM, and (OME-)Zarr containers in order to determine the most viable archival formats and their compatibility with prevalent image processing software packages. Quality loss is quantified using common full-reference metrics including SSIM[1], MS-SSIM[2], and PSNR.

We find that lossless schemes yield a maximal CR of 1.75:1, corresponding to a 43% space saving (SS). The JPEG-LS and JPEG XL codecs achieve CRs of 16.3:1 (93.9% SS) and 45.6:1 (97.8% SS) respectively at imperceptible quality loss (PSNR $\approx$ 60 dB full scale, 3D SSIM $\approx$ 0.999) on a representative 16-bit CT volume. Alongside JPEG 2000, these schemes in combination with a DICOM container[3] constitute the only lossy options widely supported among the surveyed packages. Compression times remain compatible with high-throughput acquisition, while decompression is fast enough that access times improve over the uncompressed, I/O-bound case.

To illustrate the economic implications, we estimate storage savings for the NaturArv3D collaborative project[1] between the DTU 3D Imaging Centre and the Natural History Museum of Denmark. Under realistic operating conditions, the expected savings amount to approximately 1 PB of storage annually relative to the existing data pipeline, which corresponds to an estimated annual cost reduction of 3 million DKK. This underscores that the adoption of lossy compression schemes is decisive for the economic viability of high-throughput X-ray imaging at scale.

References

[1] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image Quality Assessment: From Error Visibility to Structural Similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.

[2] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale Structural Similarity for Image Quality Assessment," in *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, Nov. 2003, p. 1398–1402 Vol.2. doi: 10.1109/ACSSC.2003.1292216.

[3] "DICOM PS3.5 2026c — Data Structures and Encoding." Accessed: Jul. 17, 2026. [Online]. Available: https://www.dicomstandard.org/current

[1] https://3dim.dtu.dk/newsarchive/2025/03/naturarv_3d

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Data / Scheme	***Small Detector*** 3072×3072×3072			*4K* 4096×4096×4096			*Large Detector* 5606×5606×4800		
	50% crop		16-bit	0% crop		16-bit	50% crop		16-bit
Uncompressed	29.0 GB	14.5 TB	756 TB	137 GB	68.7 TB	3.59 PB	151 GB	75.4 TB	3.94 PB
CR = 1	643 kr	321 kkr	16.8 Mkr	3.05 kkr	1.52 Mkr	79.5 Mkr	3.34 kkr	1.67 Mkr	87.3 Mkr
SS = 0.00%	120 kr	59.9 kkr	3.13 Mkr	568 kr	284 kkr	14.8 Mkr	623 kr	312 kkr	16.3 Mkr
	77.3 kr	38.6 kkr	2.02 Mkr	366 kr	183 kkr	9.55 Mkr	402 kr	201 kkr	10.5 Mkr
Lossless	17.1 GB	8.53 TB	445 TB	80.8 GB	40.4 TB	2.11 PB	88.7 GB	44.4 TB	2.32 PB
CR = 1.7	378 kr	189 kkr	9.87 Mkr	1.79 kkr	896 kkr	46.8 Mkr	1.97 kkr	984 kkr	51.3 Mkr
SS = 41.17%	70.5 kr	35.2 kkr	1.84 Mkr	334 kr	167 kkr	8.72 Mkr	367 kr	183 kkr	9.57 Mkr
	45.4 kr	22.7 kkr	1.19 Mkr	215 kr	108 kkr	5.62 Mkr	236 kr	118 kkr	6.17 Mkr
Lossy JPEG-LS	1.78 GB	889 GB	46.4 TB	8.43 GB	4.22 TB	220 TB	9.25 GB	4.63 TB	241 TB
level = 128	39.4 kr	19.7 kkr	1.03 Mkr	187 kr	93.5 kkr	4.88 Mkr	205 kr	103 kkr	5.35 Mkr
CR = 16.3	7.35 kr	3.68 kkr	192 kkr	34.8 kr	17.4 kkr	909 kkr	38.2 kr	19.1 kkr	998 kkr
SS = 93.86%	4.74 kr	2.37 kkr	124 kkr	22.5 kr	11.2 kkr	586 kkr	24.7 kr	12.3 kkr	643 kkr
Lossy JPEG XL	636 MB	318 GB	16.6 TB	3.01 GB	1.51 TB	78.6 TB	3.31 GB	1.65 TB	86.3 TB
$d = 0.25$	14.1 kr	7.05 kkr	368 kkr	66.8 kr	33.4 kkr	1.74 Mkr	73.4 kr	36.7 kkr	1.91 Mkr
CR = 45.6	2.63 kr	1.31 kkr	68.5 kkr	12.5 kr	6.23 kkr	325 kkr	13.7 kr	6.84 kkr	357 kkr
SS = 97.80%	1.69 kr	847 kr	44.2 kkr	8.03 kr	4.02 kkr	210 kkr	8.81 kr	4.41 kkr	230 kkr
Lossy JPEG XL	206 MB	103 GB	5.38 TB	978 MB	489 GB	25.5 TB	1.07 GB	537 GB	28.0 TB
$d = 1.0$	4.58 kr	2.29 kkr	119 kkr	21.7 kr	10.8 kkr	566 kkr	23.8 kr	11.9 kkr	621 kkr
CR = 140.5	853 mkr	426 kr	22.2 kkr	4.04 kr	2.02 kkr	105 kkr	4.44 kr	2.22 kkr	116 kkr
SS = 99.28%	550 mkr	275 kr	14.3 kkr	2.61 kr	1.30 kkr	68.0 kkr	2.86 kr	1.43 kkr	74.6 kkr

- Cell rows:
  1. Data size
  2. Storage cost — MorphSource, 3.47 USD/GB = 22.33 DKK/GB
  3. Storage cost — University of Cambridge, Research Computing Services, Research File Share
     - https://docs.hpc.cam.ac.uk/storage/prices.html
     - Backed up, 2 replicas
     - 475.00 GBP/TB/5yr = 4132.5 DKK/TB/5yr
  4. Storage cost — Backblaze B2
     - 417 USD/TB/5yr = 2.683.75 DKK/TB/yr
- Cell columns:
  1. 1 reconstruction
  2. 1 week of reconstructions (shaded, 500)
  3. 1 year of reconstructions ($500 \times 365.2425/7 \approx 26089$)
- Units have been SI-ified: mkr = milli-kroner, Mkr = mega-kroner/million, etc.

Cost estimate in abstract:

- Lossy JPEG XL with $d = 0.25$ for CR = 45.6
- Round up ***Small Detector*** estimate from ≈ 756 GB to 1 PB to account for the fact that some significant portion of samples are to be run on *Large Detector*.
- Use Cambridge RCS cost estimate, amortized over 5 years:
- $1\ \text{PB} \times 1000\frac{\text{TB}}{\text{PB}} \times 475.00\frac{£}{\text{TB} \times 5\ \text{yr}} \times 8.7\frac{\text{DKK}}{£} \times \left(1 - \frac{1}{45.6}\right) = 3\,055\,657.5\ \text{DKK}$